

Positive-Case Count and Synthetic Data Fidelity: A CART-Based Analysis of MIMIC-IV Clinical Data

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Abstract

A prior study found that in Bayesian logistic regression the stabilization threshold is governed not by sample size but by the number of positive cases for a predictor. We ask whether the same holds for CART-based synthetic data generation (`synthpop`) on MIMIC-IV ($n = 70,954$ ICU stays), using two sweeps that hold the generator fixed and vary only the input: one varies the rare stratum’s positive count k at fixed sample size, the other varies sample size n at fixed k , which together separate count from prevalence. Four measurements agree, and none of them follows sample size. Over matched eight-fold increases, raising k drives divergence from 6–17% to exactly zero at $k = 80$ in all six series and cuts interquartile width by 42–66%, while raising n moves neither: divergence stays within a few points of where it started and width changes by -19 to $+10\%$ with no consistent sign. A CART matched to the generator’s control parameters uses the comorbidity in 4–14% of replicates at every sample size, against 26–40% at $k = 250$, and a logistic fit to the same subsamples detects it in 2–16% of replicates throughout, with no consistent gain on real labels over shuffled ones. The target interactions are small, 0.050 to 0.071 on the logit scale against a creatinine slope roughly three to four times larger, and a 500-row subsample carries only 6–8% power to detect them. The mechanism therefore holds in a narrow sense but does not transfer to clinical effect sizes. Two recommendations follow: establish that a target effect is estimable in the source data before attributing any discrepancy to the generator, and calibrate variable-selection diagnostics against a permutation null, whose own rate here moves from 0.00 to 0.32 with class balance alone.

1 Introduction

Synthetic data is often proposed as a way to share clinical records without exposing patients, but whether it preserves the relationships researchers care about is a separate question from whether it looks realistic overall. A prior analysis of the `birthwt` dataset found that under Bayesian logistic regression, posterior stability depends on the number of positive cases for a predictor rather than on overall sample size.

This report asks whether the same mechanism holds for CART-based sequential synthesis. The generator is held fixed throughout; the question is not which tree-based method performs best but how one of them behaves when the stratum it must reproduce is small. Because positive count and prevalence move together whenever sample size is fixed, a single sweep cannot attribute a change to either, so two complementary sweeps are run. Code is available at [GitHub Repository](#).

2 Data

2.1 Cohort

MIMIC-IV v2.2 [3], restricted to hospital admissions with an associated ICU stay. One row per ICU stay (73,181 stays) was built from `icustays`, `patients`, `admissions`, and `diagnoses_icd`, the last supplying the comorbidity flags. Dropping the 3% of stays with no first-day creatinine leaves 70,954 rows; all subsampling draws from this pool.

2.2 Comorbidity Screening

Five comorbidities were defined by simplified ICD-9/ICD-10 groupings and counted by admission rather than by ICU stay, the unit the pool is indexed on: COPD 7,244, CHF 17,412, Diabetes 19,806, Renal Failure 25,369 and Hypertension 34,979. The three least common were carried forward. All five are common in the pool; the rare stratum is created by the subsampling below, not found in the data.

2.3 Continuous Variable Selection

Creatinine was chosen from five first-day lab candidates for its 97% coverage and its link to CHF through cardiorenal syndrome. The maximum in the first 24 hours was taken, capped at the 99.5th percentile (11 mg/dL).

3 Methods

3.1 Experimental Design

Both sweeps draw stratified subsamples, synthesize each with `synthpop::syn()` [2], and fit

$$\text{hospital_expire_flag} \sim \text{age} + \text{creatinine} + \text{comorbidity} + \text{comorbidity}:\text{creatinine}$$

to both the real subsample and the synthetic table, giving paired estimates $\hat{\beta}_{\text{real}}$ and $\hat{\beta}_{\text{syn}}$ of the interaction, at $M = 100$ Monte Carlo replicates per grid point.

Experiment	Held fixed	Varied	What moves
1. Count sweep	$n = 500$	$k \in \{5, 10, 20, 40, 80, 150, 250\}$	count and prevalence together
2. Size sweep	$k = 20$	$n \in \{250, 500, 1000, 2000, 4000\}$	prevalence only, 8% to 0.5%

3.2 Analyses

We conduct three analyses. (1) A full-pool fit of the interaction supplies the reference every subsample estimate is read against. (2) The two sweeps compare $\hat{\beta}_{\text{real}}$ against $\hat{\beta}_{\text{syn}}$ at each grid point on the median and interquartile width of the converged estimates, with the divergence and inestimable rates kept separate; Section 5 defines the two ways a fit fails. (3) A tree diagnosis refits the subsamples both sweeps draw with `rpart` under `synthpop::syn.cart`'s control parameters and records whether the comorbidity earns a split, against a likelihood-ratio test on the same rows that drops the comorbidity and its interaction together, so both are asked the same question. Every configuration is re-run with the comorbidity column shuffled, which leaves the positive count intact and destroys its association with the outcome, supplying the null.

4 The Full-Pool Reference

MIMIC-IV carries no known ground truth, so the interaction fit on the whole pool stands in as the reference, $\hat{\beta}_{\text{full}}$, and every subsample estimate below is read against it. The three interactions differ in sign, CHF and Diabetes dampening creatinine's main effect while COPD steepens it, and all three are roughly three to four times smaller than that main effect.

Comorbidity	Creatinine	$\hat{\beta}_{\text{full}}$	SE	p
CHF	0.208	-0.0497	0.0128	1.1×10^{-4}
Diabetes	0.234	-0.0709	0.0126	2.0×10^{-8}
COPD	0.188	+0.0668	0.0197	6.9×10^{-4}

Table 1: Comorbidity-creatinine interaction on the full pool, with the creatinine slope it modulates.

5 Estimation Failure

5.1 Two Failure Modes

Fits at the smallest counts failed often enough to matter, and the failures turned out to be of two distinct kinds. We report their rates as a result rather than a filtering step: whether an estimate can be produced at all separates the two levers more sharply than the spread of those that succeed. Every quantile that follows is computed on converged fits only.

A fit is **inestimable** when the table holds no positives and the interaction term comes back as NA; **divergent** when the term is estimated but runs past $|2|$, marking a likelihood maximized at the boundary [1] rather than a real effect. Only divergence appears in the real arm, which is built to hold exactly k positives; the synthetic table generates its own column and can produce none.

The cutoff follows the scale of the target, around 0.05 in the full pool, and recomputing at 5 and 10 leaves every comparison pointing the same way. Both modes are reported as shares of all M replicates, so converged, inestimable and divergent sum to one.

5.2 Failure Under Each Lever

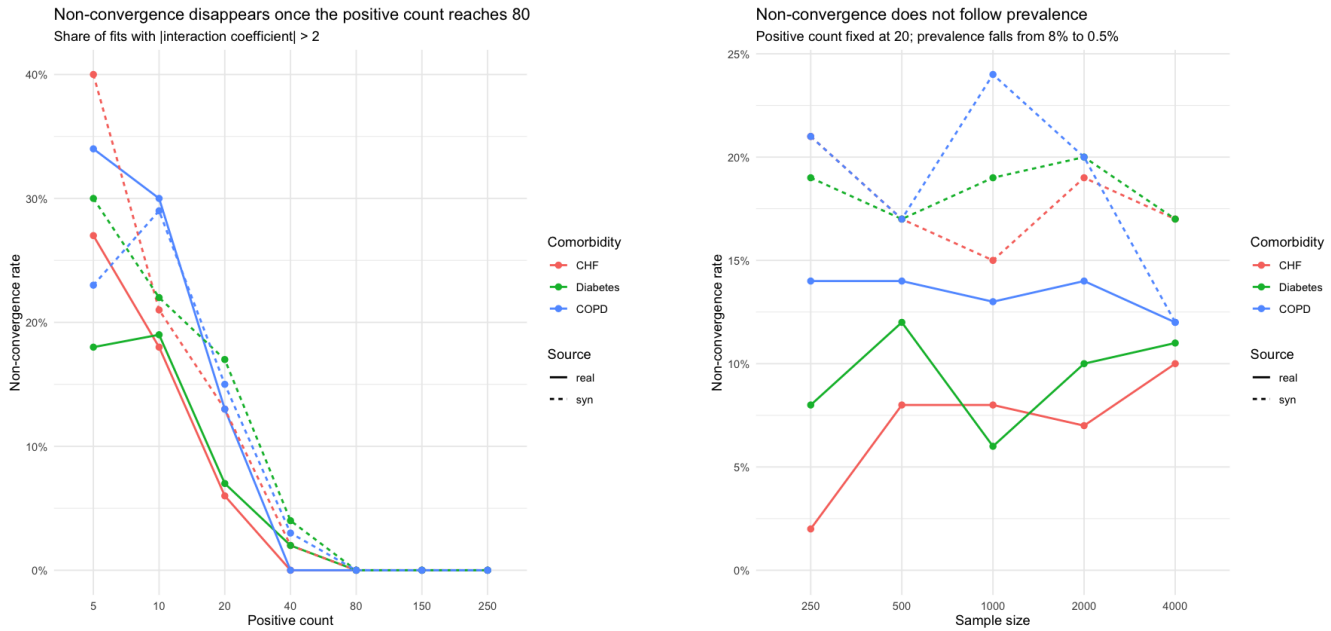


Figure 1: Divergence rate under each lever, three comorbidities $\times$ two sources in both panels. Left: positive count varying at $n = 500$. Right: sample size varying at $k = 20$, where prevalence falls from 8% to 0.5%. Inestimable fits occur on the synthetic side only and are reported in the text.

Count sweep. All six curves reach zero at $k = 80$ and stay there. The threshold moves with neither prevalence, which varies nearly threefold across the three comorbidities, nor the sign of the interaction. Below it, divergence runs at 18–40% at $k = 5$ and is higher on the synthetic side in ten of the twelve failing cells. The inestimable rate is narrower still: synthetic only, $k = 5$ only, at 4–6%.

Size sweep. No sample size does the same. Over the matched 8-fold increase in n every series changes by between -29% and $+25\%$, and none comes near zero (Figure 1, right). COPD is the plainest case: its real arm fails at 12–14% at every one of the five sample sizes. The inestimable rate is zero in all fifteen cells, including at $n = 2000$, where prevalence matches the $k = 5$ cells that lost the stratum 4–6% of the time. Failure follows the count, and only the count.

6 Results

6.1 Experiment 1: Count Sweep

Spread narrows steadily with the count: from $k = 20$ to $k = 250$ every one of the six series loses 55% to 69% of its width, and none has flattened by the end of the range. It is still not enough. Even the narrowest cell, the CHF real arm at $k = 250$, has an interquartile width of 0.194 against its reference of 0.0497 (Table 1), nearly four times the quantity being estimated. The low end of the span rests on fewer replicates, 6% to 44% being discarded below $k = 40$, but all of them for diverging, so the decline is if anything understated.

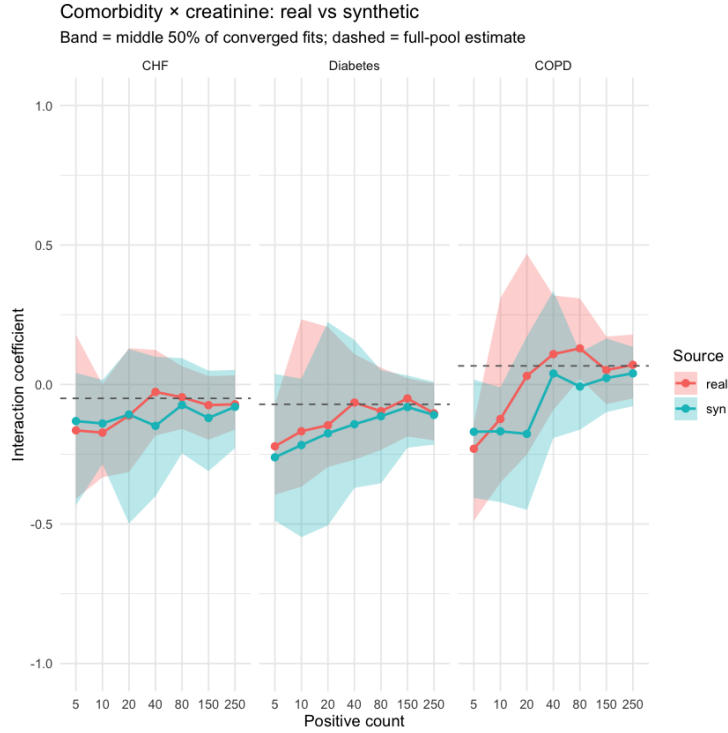


Figure 2: Comorbidity $\times$ creatinine, real vs. synthetic. Band = middle 50% of converged fits, dashed line = $\hat{\beta}_{\text{full}}$.

What low counts cost is precision, not location: in 39 of the 42 cells the middle half of the estimates still contains $\hat{\beta}_{\text{full}}$, the three exceptions all COPD at $k \leq 10$. The two arms differ on both counts, and in the same direction. The synthetic median sits below the real one in 17 of the 21 counts, and where neither side fails at all ($k \geq 80$) synthetic widths run 1.1 to 1.6 times the real ones for CHF and Diabetes and roughly equal for COPD.

6.2 Experiment 2: Size Sweep

The narrowing Experiment 1 found does not reappear under the other lever. Over roughly eightfold increases in each, k from 20 to 150 and n from 500 to 4000, the count cuts width by 42% to 66% in all six series while the sample size changes it by between -19% and $+10\%$ with no consistent sign (Figure 3). Across the full sixteen-fold range of n the widths never leave the band Experiment 1 recorded at $k = 20$. The count was held at 20, the last configuration in which it still binds: failure there runs 6–17% of replicates against 0–4% at $k = 40$ and none from $k = 80$ up. If sample size can substitute for count, this is where it would show.

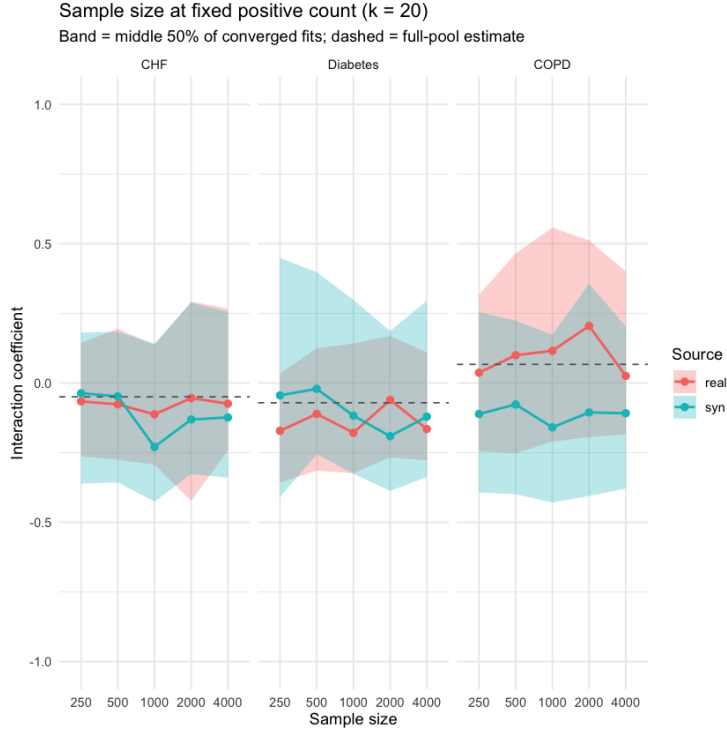


Figure 3: Comorbidity $\times$ creatinine with the positive count held at 20. Band = middle 50% of converged fits, dashed line = $\hat{\beta}_{\text{full}}$. In the COPD panel the synthetic median sits below the reference at every n and the real median above it.

Width is not the only thing sample size fails to fix. COPD is the one comorbidity whose interaction is positive, $+0.0668$, and Experiment 1 recovers that sign once the count passes 20. Holding the count at 20 does not: the synthetic median is negative at all five sample sizes, from -0.077 to -0.159 , matching the -0.177 Experiment 1 recorded at the same count. Sixteen times the data leaves the sign where twenty positives put it.

The size effect itself cannot be resolved from noise. The configuration $k = 20$, $n = 500$ appears in both experiments and so is measured twice under independent seeds; the readings differ by 13%, 10% and 0.3%, the same order as the variation across n .

6.3 Tree Diagnosis: The Diagnostic Needs a Null

Count sweep

What looks like recovery is class balance, not signal. Read without a control the diagnostic rises from 0 at $k = 5$ to 0.26–0.40 at $k = 250$, but the permutation null rises with it, from 0.00 to as high as 0.32 (Table 2). What grows alongside the count is the balance of the column itself, from 1% positive at $k = 5$ to 50% at $k = 250$, and a balanced binary variable is structurally easy to split on whether or not it carries signal [4]. Any diagnostic of this form requires permutation calibration before it can be read.

k	CHF		Diabetes		COPD	
	Observed	Null	Observed	Null	Observed	Null
5	0.00	0.00	0.00	0.00	0.01	0.00
10	0.04	0.01	0.00	0.01	0.04	0.01
20	0.08	0.01	0.02	0.04	0.09	0.05
40	0.21	0.07	0.15	0.08	0.16	0.07
80	0.21	0.23	0.27	0.15	0.30	0.19
150	0.28	0.28	0.28	0.32	0.28	0.21
250	0.27	0.30	0.40	0.20	0.26	0.27

Table 2: Proportion of replicates in which CART selects the comorbidity as a split, against its permutation null.

Cell by cell the diagnostic is underpowered. Paired tests over the 100 replicates clear $p < 0.01$ in two of the 21 cells, CHF at $k = 40$ ($p = 0.004$) and Diabetes at $k = 250$ ($p = 0.002$), and only the latter survives a Bonferroni correction; both sit isolated among non-significant neighbours. Pooled over the grid the real column does beat the shuffled one, 249 discordant replicates to 164, but the margin is thin because the null has risen to meet it.

Size sweep

Holding the count at 20 runs class balance the other way, from 8% positive at $n = 250$ to 0.5% at $n = 4000$, and the null falls with it, ending between 0.00 and 0.03 (Table 3). The null therefore tracks the balance of the column and nothing else, in a range the count sweep never reached. The observed rate does not follow it down, staying between 0.04 and 0.14 at every size, where the count sweep left it at $k = 20$.

n	Prevalence	CHF		Diabetes		COPD	
		Observed	Null	Observed	Null	Observed	Null
250	8.0%	0.05	0.03	0.07	0.02	0.06	0.08
500	4.0%	0.04	0.02	0.08	0.04	0.06	0.01
1000	2.0%	0.14	0.02	0.07	0.02	0.08	0.04
2000	1.0%	0.06	0.01	0.12	0.02	0.06	0.02
4000	0.5%	0.07	0.00	0.09	0.01	0.04	0.03

Table 3: CART’s use of the comorbidity against its permutation null, with the positive count held at 20.

With the null near zero the signal becomes legible. Pooled over this grid the real column wins 104 discordant replicates to 32, and 14 of the 15 cells point the same way, against 249 to 164 in the count sweep. Per cell the tests stay underpowered, only CHF at $n = 1000$ surviving a Bonferroni correction over 15 comparisons, but the direction is no longer in doubt: the diagnostic is readable where its null is near zero, which on this data means low prevalence.

Neither sweep implicates CART on its own. A GLM fitted to the same subsamples detects the comorbidity in 2–16% of replicates in every cell of both sweeps, against the 6–8% predicted by rescaling the full-pool standard errors to $n = 500$, and it does no better on real labels than on shuffled ones: over the size sweep the observed rate averages 9.1% and the permuted 9.5%, with no trend in n . Both methods are blind at these sizes; CART is the one that fails visibly, by declining to spend a split.

7 Discussion

Two failure modes, not one. Reporting the failure rates separately from the spread changed the reading of this data more than any other decision. Mixing them inflates apparent instability at low counts, since divergent fits push the quantiles outward, and it hides the sharpest result in the study, the common threshold at $k = 80$.

A privacy-fidelity tension. CART’s reluctance to isolate a subgroup with few supporting cases is not simply a limitation to work around. A tree willing to carve out a node of five patients would also be more likely to overfit to, and risk re-identifying, those individuals, and that same reluctance is what leaves the stratum missing from the synthetic table in the inestimable cases at $k = 5$. The tension is a property of the method rather than an implementation defect, though only its fidelity side is measured here.

The design, not the generator. A 500-row subsample carries 6–8% power against these interactions, and the GLM arm returned 6.7–10.1%. Theory and experiment agreeing on a null result is what rules out a defect in the pipeline and locates the failure in the design rather than in the generator.

Limitations. The diagnostic tree is a proxy for the generator: three predictors rather than the full feature set `synthpop` synthesizes jointly, matched on control parameters but not on its sequential visit order, and counting the comorbidity as used wherever it appears rather than requiring it beneath a creatinine split. All three are permissive, which makes non-selection the more reliable of the two outcomes. Monte Carlo error at $M = 100$ reaches 13% on interquartile widths, and the paired tree tests turn on at most 40 discordant replicates per cell in the count sweep and 14 in the size sweep, so a cell that misses significance is not evidence that nothing is there. One continuous variable and one outcome were examined, Renal Failure and Hypertension were screened but not analyzed, and all results come from a single MIMIC-IV snapshot.

8 Conclusion

The positive count governs every quantity measured here, and sample size governs none of them. Divergence falls to zero at a count of eighty for all three comorbidities and both data sources; interquartile width falls by 42–66% over a matched increase in the count; CART’s use of the comorbidity is flat across a sixteen-fold change in n ; and a logistic fit to the same subsamples shows no consistent separation from its permutation null under either lever. Four readings, two of them from data the synthesizer never touched. In that narrow sense the finding from Bayesian logistic regression does extend to CART-based synthesis.

The wider claim does not survive contact with clinical effect sizes. At practical subsample sizes neither arm recovers the MIMIC-IV interactions: even the narrowest cell spans nearly four times the quantity being estimated, and with the count held at 20 the COPD synthetic median carries the wrong sign at every sample size tested. Where the two can be compared, the synthetic side is worse on both failure and spread.

For synthetic data evaluation, two things follow: establish that the target effect is estimable in the source data before attributing a discrepancy to the generator, and calibrate any variable-selection diagnostic against a permutation null, since its null moves with class balance from zero to 0.32 across the range tested here.

References

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